An In-Depth Analysis and Application of GDRNPP for Monocular 6D Object Pose Estimation

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Abstract

Augmented reality (AR) overlays have the potential to revolutionize surgical techniques by providing surgeons with real-time visualization of anatomical information, surgical instrumentation, and procedural guidance. However, accurately registering virtual objects to the real scene is a significant challenge that is essential for effective AR applications in surgery. This paper proposes a novel framework that incorporates the Geometry-Guided Direct Regression Network (GDR-Net(2)), and its extension GDRNPP (1), into an endoscope-based AR applications. GDRNPP is a robust and efficient registration method that utilizes local image features and depth information to achieve accurate alignment. Our extension adapts GDRNPP to a virtual scene, enabling the seamless integration of virtual objects into the endoscope's field of view.

Experimental results demonstrate the effectiveness of our approach in achieving real-time and sustained registration for AR overlay in surgical robotics scenarios. Our method has the potential to improve the accuracy, safety, and efficiency of surgical procedures.

1. Introduction

In the realm of surgical interventions, augmented reality (AR) has emerged as a transformative technology with the potential to revolutionize surgical visualization and guidance. By seamlessly overlaying real-time anatomical information, surgical instrumentation, and procedural guidance onto the surgeon's field of view, AR holds immense promise for enhancing surgical precision, safety, and efficiency. However, the effectiveness of AR-based surgical guidance systems is fundamentally intertwined with the accuracy and real-time performance of 6D object pose estimation. This crucial task involves determining both the 3D location and orientation of an object in space, enabling the precise registration of virtual objects onto the real-world scene.

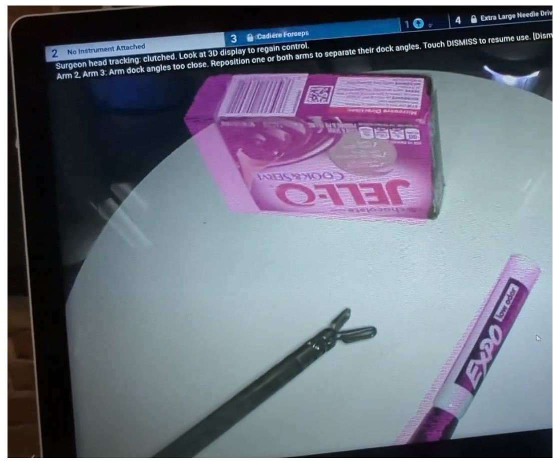

Figure 1. **Demo of Endoscopic Overlay for rigid object registration.** We automatically localize objects of interest in the surgical scene and overlay them on the 3D surgeon console display.

Conventional 6D object pose estimation methods can be broadly categorized into two main approaches: indirect methods and direct methods. Indirect methods establish 2D-3D correspondences between points in the image and a 3D model of the object, followed by solving a Perspective-n-Point (PnP) problem to determine the object's pose. While these methods generally offer higher accuracy, they are computationally expensive, making them less suitable for real-time applications like AR-based surgical guidance systems. Direct methods, on the other hand, directly regress the pose parameters from image features, offering significantly faster processing speeds and improved computational efficiency. However, they typically come at the cost of reduced accuracy compared to indirect methods.

Recent advancements in deep learning have led to the development of new direct methods that achieve competi-

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tive accuracy while maintaining real-time performance. One such method is GDRNPP, which emerged victorious in the BOP Challenge 2022(3). GDRNPP employs geometry-guided direct regression, point-pair features, and a novel post-processing strategy to achieve accurate and efficient pose estimation.

This paper proposes a proof-of-concept framework for an AR-based surgical guidance system that leverages the GDRNPP algorithm for real-time 6D pose estimation of rigid objects seen from a stereo endoscope. The estimated poses are utilized to overlay the corresponding 3D models over the physical objects. This real-time overlay of virtual objects, when applied to context-relevant objects, provides surgeons with enhanced visualization and guidance, potentially leading to improved surgical precision and safety.

2. Related Work

Our work builds upon two areas of active research: 6D object pose estimation and Augmented Reality (AR) applications in surgical robotics.

2.1. BOP Challenge 2022

Estimating the 6 Degrees of Freedom (DoF) pose of rigid objects from RGB or RGB-D images is crucial for various computer vision tasks. The Benchmark for 6D Object Pose Estimation (BOP) (3) serves as a key platform for evaluating state-of-the-art approaches. It provides a standardized evaluation methodology, online evaluation system, and twelve datasets featuring nearly 200 texture-mapped 3D models, over 700K training RGB-D images, and over 100K test RGB-D images with ground-truth 6D object poses.

The BOP Challenge 2022, the fourth iteration in the series of public competitions, attracted 49 pose estimation methods (3). Among them, the GDRNPP algorithm emerged as the top performer, achieving an average recall of 83.7% (1). This average recall is calculated as the average error in the estimated pose with regard to the ground truth pose using Visible Surface Discrepancy (VSD), Maximum Symmetry-Aware Surface Distance (MSSD), and Maximum Symmetry-Aware Projection Distance (MSPD). Our work utilizes this high-performing algorithm for accurate pose estimation in the context of surgical robotics.

2.2. AR for Surgical Robotics

AR technology has found promising applications in surgical robotics, offering functionalities such as image overlay, surgical navigation, and surgical simulation(15).

Image overlay methods project pre-operative or intra-operative images (e.g., CT, MRI, ultrasound) onto the patient's body or the surgeon's field of view, allowing visualization of internal structures without requiring large incisions. Surgical navigation methods utilize AR to display the position and orientation of surgical instruments (e.g., endoscopes, needles, robots) relative to the patient's anatomy, facilitating precise and safe manipulation (16; 17; 18; 19).

Despite their benefits, existing methods often suffer from limitations:

- **Low accuracy**: Some methods struggle with accurate pose estimation, potentially impacting their effectiveness.
- **High latency**: Real-time applications require low latency to ensure smooth interaction and avoid disruption of the surgical workflow.
- **Complex calibration**: Many methods necessitate intricate calibration procedures, adding complexity and potential for error.
- **Limited applicability**: Certain methods are specific to certain types of instruments, environments, or tasks, limiting their overall usefulness.

Furthermore, existing methods often rely on external markers or sensors for registration, require separate displays for AR information, or lack realism in simulation environments. These limitations hinder the adoption and effectiveness of AR in surgical applications.

Our framework has several advantages over the existing methods, such as high accuracy, low latency, simple calibration, and broad applicability. Our framework does not require any external markers or sensors, as it uses the GDRNPP algorithm to estimate the pose of the objects from a single image, reducing the complexity and error of the registration process. Our framework does not require any separate display or device, as it uses the Unity game engine to generate the AR overlay on the same display that shows the endoscopic image, enhancing the immersion and convenience of the surgeon. Our framework can be applied to various surgical scenarios, such as laparoscopic, open, or robotic surgeries, as it can handle different types of objects, lighting conditions, and camera poses, demonstrating the robustness and versatility of our framework.

3. Methods

This paper proposes a novel framework for 6D pose estimation and augmented reality (AR) visualization during robotic-assisted surgery. The framework consists of three main components that work together to enhance the surgeon's situational awareness and facilitate improved surgical performance.

The first component of the framework is a 6D pose estimation algorithm that accurately determines the pose of trained objects in 3D space from images captured by the stereo endoscope. Based on the results of the BOP 2022 Challenge, we have selected the GDRNPP method.

The 6D pose estimation results are seamlessly integrated with the surgical robot control system and the stereo endoscope to provide the surgeon with a real-time overlay of the objects' position and orientation within the surgical scene. The integration process involves the creation of an endoscope digital twin. This is achieved using the Unity game engine along with stereo camera calibration and robot forward kinematics. We employ a communication protocol that enables real-time data transfer and synchronization for sending GDRNPP pose estimates to Unity

The augmented reality overlay generated by Unity is superimposed onto the 3D surgeon console display, providing the surgeon with a unified visualization of the real and virtual worlds. For this project, that involves overlaying a 3D-scanned model on top of its physical counterpart.

(a) Raw Endoscope Image

(b) Deinterlaced and Rectified Image

Figure 2. **Deinterlacing and rectification of endoscope images** (a) The raw stereo endoscope image is horizontally interlaced with the left and right camera. (b) For input into GDR-Net, the image must be deinterlaced, rectified, cropped, and scaled to match the training data.

3.1. Endoscopic 6DoF Pose Estimation using GDRNPP

Building upon the foundation of GDR-Net, GDRNPP introduces a novel patch-based framework that leverages dense correspondence matching, surface region attention, and iterative refinement to achieve accurate and real-time 6D object pose estimation. With a few modifications, this algorithm can work with 3D endoscopic images to get 6D pose of objects seen in the surgical field.

3.1.1. Image Preprocessing

Endoscopic images are horizontally interlaced, alternating between left and right camera, such that the surgeon (wearing passive polarized glasses) can perceive 3D depth information on the surgeon monitor. These images are also significantly distorted so that the surgeon can perceive a wide field of view within the body cavity. To obtain rectified stereo pairs, we first de-interlace the images and then apply OpenCV calibration parameters. We then resize the original 1920x1080 images to 640x480, which is the size of the training data images. This ensures that the GDR-Net ConvNext backbone network can efficiently extract high-level features representing the object's appearance. Subsequently, the pixel values are normalized to lie within a standard range, such as [0, 1], to handle differing image intensities. This normalization helps to reduce the impact of varying image intensities, improving the robustness of the pose estimation process. This processing is showcased in Figure 2, which highlights the difference between the raw endoscope image input and the processed image needed for input to the GDR-Net.

3.1.2. Object Detection and Region of Interest

An off-the-shelf object detection algorithm, in this case YOLOX, is employed to detect the bounding box of the object of interest in the image. This step is crucial for identifying the object of interest and extracting a RoI that contains the object. The RoI is then passed to the subsequent processing steps for more focused processing.

3.1.3. Feature Extraction and Mask Prediction

The cropped RoI is passed through the GDR-Net ConvNext backbone network. This deep learning model is trained to extract high-level features from images, capturing the essential geometric and visual information crucial for pose estimation. The extracted features are then utilized to predict two masks: an amodal mask and a visible mask. The amodal mask represents the full extent of the object, including occluded parts, while the visible mask indicates the portions of the object that are visible in the image.

3.1.4. Dense Correspondence Matching

Dense correspondences between points on the object's surface and points in the image are established using the predicted amodal mask. Dense correspondence matching is a challenging task, especially in the presence of occlusions and complex backgrounds. However, the predicted amodal mask can be used to guide the correspondence matching process, improving the accuracy and robustness of the matching. The established correspondences are then utilized to generate a dense point cloud representing the object's surface.

3.1.5. Surface Region Attention

A surface region attention module is applied to the dense point cloud to generate a weight map. This module leverages the geometric cues provided by the dense point cloud to learn a weight map that assigns higher weights to points that are more likely to be informative for pose estimation. This helps to suppress noisy or irrelevant points in the point cloud, directing attention to the most informative regions for accurate pose determination.

3.1.6. Patch-PnP Algorithm

The RoI is divided into patches. For each patch, the Patch-PnP algorithm is employed to estimate its 6D pose using the weighted point cloud obtained from the Surface Region Attention. This patch-based approach provides initial pose estimates for various parts of the object. These initial pose estimates are then integrated to generate a more accurate global pose estimate for the object. This final pose is outputted in the form of a 4x4 transformation matrix.

3.2. Virtual Scene Pose Reconstruction for AR Overlay

We generate 3D endoscopic overlays using the Unity game engine. We use the DDS Communication protocol to establish a real-time network connection between the Python GDRNPP environment and the Unity environment. We transmit the per-object transformation matrices from the GDRNPP output into Unity, convert them to the appropriate coordinate system, and update the transformation properties of the corresponding 3D models in Unity according to a cached camera pose that matches the endoscopic image capture. This ensures a consistent and accurate visualization of the endoscopic scene. Figure 3 shows a proof-of-concept of this method using a webcam to capture pre-trained objects.

This Unity scene is rendered on top of the endoscope image on the surgeon display, resulting in an augmented reality visualization showing the virtual objects localized and overlaid on their physical counterparts.

3.3. YCB-V Dataset

The YCB-V dataset was selected due to its availability and variety of objects. It has been well-documented and was included in the BOP 2022 Challenge. The dataset provides detailed annotations, including 6D object poses and pixel-wise segmentation, essential for training our model with the precision required in medical applications. Furthermore, the inclusion of realistic video sequences with varying lighting and object arrangements simulates the complex scenarios encountered in surgeries, ensuring our model's robustness and adaptability.

3.4. Model Optimizations

The motivation to optimize the model for implementation on stand-alone Augmented Reality devices, such as Hololens2 and smartphones, is driven by the computational limitations of these platforms. These devices offer limited computing power compared to traditional environments, necessitating efficient model operation within these constraints to avoid lag in inference speed, crucial for AR applications where real-time processing is key.

A primary objective includes ensuring cross-platform and cross-framework compatibility, broadening the model's usability and enhancing accessibility. Additionally, improving the model's accuracy and robustness is vital for AR applications, where interactions with the real world are overlaid with digital information. Resource optimization, aimed at reducing computational resource usage, power consumption, and heat generation, is also significant for handheld and wearable devices.

To achieve these objectives, two key experiments were conducted:

3.4.1. ONNX Conversion

The conversion process to ONNX format aimed at enhancing model versatility and efficiency. ONNX facilitates model interoperability across different frameworks and hardware platforms, beneficial for varied deployment environments. This conversion supports advanced optimization techniques for improved inference speed without compromising accuracy.

3.4.2. Model Optimization

The optimization phase focused on increasing inference speed, reducing memory footprint, and lowering power consumption. The conversion from FP32 to INT8 precision was pivotal, making the model suitable for edge computing devices and efficient for battery-powered devices. Tailoring the model to leverage hardware accelerators like GPUs, TPUs, and ASICs was also a focus to enhance performance.

4. Results

We conducted a series of experiments to demonstrate the performance and effectiveness of our proposed framework for 6D pose estimation and augmented reality visualization in surgical scenarios. We used the YCB-V dataset as the training and testing data for the GDRNPP algorithm,

(a) Objects detected in image

(b) Unity reconstruction

Figure 3. **Visualization of the 6D pose estimation results using the GDRNPP algorithm and the Unity game engine.** (a) The raw image with the detected objects and their bounding boxes. (b) The virtual representation of the objects in Unity, where the relative 6D poses are applied to each object according to the Unity camera. The visualization shows the accuracy of the pose estimation and the consistency of the scene.

and we used a stereo endoscope to capture real-world images of of some of those pre-trained rigid objects such as a Expo Marker, banana, and Jell-O pudding box. We also implemented our framework on a Unity game engine and integrated it with the surgical robot control system and the stereo endoscope. We qualitatively evaluated our framework using various scenarios and validated the effectiveness of this methodology. We also demonstrated the feasibility and utility of our framework for providing real-time and accurate AR overlays during surgical procedures.

We expand upon these results by performing preliminary model optimization procedures, such as model framework conversion to facilitate performance improvements and hardware interoperability.

4.1. Qualitative Evaluation of Endoscopic 6D Pose Estimation and Overlay

We qualitatively evaluated the 6D pose estimation using the endoscopic images captured by the stereo endoscope. Figure 4 shows some examples of the 6D pose estimation results on the endoscopic images. Our method showed a high degree of accuracy and robustness in estimating the 6D pose of our selected objects.

Figure 5 showcases the application of these results to an endoscopic AR overlay. Our method showed a high degree of accuracy and consistency in aligning the virtual object and the physical object in the endoscopic image. Our method also showed a good performance in adapting to different camera poses and different object arrangements, ensuring a seamless integration of the virtual and real worlds.

4.2. ONNX Conversion Results

The conversion to ONNX showed performance improvements:

On Nvidia 3060 GPU: The performance improved from 0.12 seconds per frame (baseline) to 0.11 seconds per frame post-conversion.

On CPU: The performance improved from 6.0 seconds per frame (baseline) to 5.2 seconds per frame post-conversion.

Table 1 details these results. Importantly, the model's accuracy remained consistent with the original PyTorch version during this process.

	On Nvidia 3060	**On CPU**
Baseline Validation	0.12	6.0
ONNX Validation	0.11	5.2

Table 1. Model Inference Speed Per Frame

4.3. Model Optimization Results

The conversion from FP32 to INT8 precision demonstrated significant improvements in inference speed, memory usage, power consumption, and hardware utilization, as shown in Table 2. These results indicate the model's enhanced efficiency and broader deployment capabilities.

	On Nvidia 3060	**On CPU**	**Model Size**
FP32 ONNX	0.11	5.2	375.7 MB
INT8 ONNX	0.09	3.2	101.9 MB

Table 2. Optimized Model Inference Speeds Per Frame

Figure 4. **Examples of 6D pose estimation reconstruction from endoscopic images.** The GDRNPP method of 6D pose estimation is effectively able to reconstruct the camera-relative pose from endoscopic images. The top images are the YOLO-X object detection with bounding box visualizations. The bottom images are the Overlay pose reconstruction using the textured 3D object models.

5. Future Work

We have demonstrated a novel framework for 6D pose estimation and augmented reality (AR) visualization in surgical scenarios using the GDRNPP algorithm. Our framework can provide real-time and accurate AR guidance and education during surgical procedures, potentially improving the precision and safety of surgical interventions. However, our framework can be further improved in several aspects, such as:

- **Hardware acceleration**: We can speed up the inference time of our framework by using more powerful GPU hardware, such as RTX 3080, which can reduce the latency from 3.8s to 0.7s per frame.

- **Model optimization**: We can increase the efficiency and effectiveness of the GDRNPP algorithm by training it on a smaller and more relevant subset of objects, such as surgical instruments, instead of using the pre-trained model on the YCB-V dataset, which contains 21 objects.

- **Depth refinement**: We can enhance the accuracy of the 6D pose estimation and the AR visualization by incorporating depth refinement techniques, such as CIR (**?**) or RAFT-Stereo (14), which can refine the depth estimation and the pose estimation of our framework.

- **Realistic evaluation**: We can validate the utility and efficacy of our framework by evaluating it on a more realistic and challenging dataset that contains context-relevant objects and scenarios for surgical applications, such as laparoscopic or open surgeries.

In addition to these improvements, we also envision several directions for future development of 6D pose estimation technology, such as:

- **Standalone AR implementation**: We aim to implement our optimized 6DoF pose estimation model on standalone AR devices, such as Hololens2 or smartphones, which can provide surgeons with seamless and intuitive AR overlays during surgical procedures, without the need for external hardware or software

- **Model robustness**: We plan to improve the robustness of our model to handle varying lighting conditions, which are common in dynamic environments such as operating rooms. We intend to develop algorithms that can adapt to the changes in shadows, reflections, and contrasts, and maintain high accuracy in real-time.

- **Multi-modal data fusion**: We aspire to expand our model to include multi-modal data fusion, which can integrate various data types, such as visual, infrared, depth, and ultrasonic sensors, to achieve a more comprehensive understanding of the environment. This

Figure 5. **Examples of endoscopic AR overlay visualization results from 6D Pose Estimation** The raw endoscopic images are overlaid with a virtual scene in which the virtual object is textured with a transparent pink material. Its 6D pose is assigned to its virtual camera in the same manner as the 6D pose estimate is acquired relative to the endoscope camera.

can not only improve the pose estimation accuracy, but also make the model more resilient to sensor-specific noise or failure. Moreover, this can enable the fusion of visual data with haptic feedback, which can give surgeons a more natural feel for their tools and the tissues they interact with.

These directions represent some of the potential applications and extensions of 6D pose estimation technology, which can benefit not only surgical practices, but also other domains, such as robotics, autonomous navigation, and interactive education, among others. We believe that our framework and future work can push the boundaries of what is currently achievable, and pave the way for groundbreaking innovations that will transform society at large

6. Conclusion

This project has successfully showcased the integration of the GDRNPP algorithm for enhanced 6D pose estimation in augmented reality (AR) environments, targeting surgical precision and operational safety. Our method, which adeptly aligns virtual and physical spaces within an endoscope's viewpoint, has been rigorously tested for accuracy, even amidst the computational constraints of AR devices. Crucially, the project ventured into model optimization, conducting experiments that not only preserved the real-time capabilities necessary for surgical procedures but also improved inference speeds and reduced computational demands. These advancements solidify the framework's potential for application in standalone AR surgical devices, marking a significant step towards the next frontier in AR-assisted surgeries and paving the way for future innovations that could redefine surgical practices and patient care.

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